Status	Finished
Started	Wednesday, 14 October 2026, 9:11 PM
Completed	Wednesday, 14 October 2026, 9:14 PM
Duration	2 mins 58 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Combining sorted sublists
c. Dividing the input into smaller subproblems
d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Insertion Sort
b. Bubble Sort
c. Merge Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It recursively sorts subproblems
b. It divides the list into sublists
c. It bubbles up the largest element in each pass
d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Heap Sort
b. Merge Sort
c. Bubble Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **5**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It makes it easier to select items based on their relationship to the rest of the items
c. It complicates the selection of items
d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Merge Sort
b. Binary Search
c. Bubble Sort
d. Linear Search

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Merge Sort
c. Bubble Sort
d. Binary Search

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting is rarely used in programming
b. Sorting helps in finding duplicates quickly
c. Sorting decreases the efficiency of selection operations
d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **9**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting
b. Merging
c. Complexity
d. Searching

The correct answer is:
Complexity

Question **10**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Rearranging

b. Searching

c. Merging

d. Sorting

The correct answer is: Sorting

Question **11**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. order()

b. arrange()

c. sort()

d. sorted()

The correct answer is: sort()

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Linear Search

b. Binary Search

c. Bubble Sort

d. Merge Sort

The correct answer is: Bubble Sort

Question **13**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It complicates the analysis process
b. It allows for quicker identification of duplicates
c. It has no effect on the analysis process
d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It allows for efficient parallel processing and sorting of data
b. It simplifies the sorting process by using only one pass
c. It only works on small datasets
d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **15**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Deleting data from a list
b. Inserting data into a list
c. Searching for data in a list
d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format